

Asher Erickson
Colin Truran
Elise Longenecker
Yvan Longin

Applied Analytics Project - Week 10 Assignment

Data Improvements and Enhancements:

In Week 10, we focused on improving the dataset used in Week 9 by implementing three new data enhancements. These improvements were applied across the training, validation, and test datasets.

The first improvement was the removal of near-duplicate features. Highly correlated variables with a correlation close to 1 were identified and removed. These redundant features do not add new information to the model and can increase noise and makes the process more complex than necessary. By eliminating them, the model is able to focus on more meaningful signals.

The second improvement was handling outliers through quantile clipping. For each feature, the 1st and 99th percentiles were calculated using only the training dataset. Values outside this range were clipped across all datasets. This reduces the impact of extreme values that can distort model learning which can happen in tree-based models like XGBoost.

The third improvement was the addition of new engineered features based on matchup context. These included absolute seed differences, absolute efficiency margin differences, and indicator variables for “toss-up” games and moderately competitive matchups. These features were designed to better capture the relative strength between teams and highlight difficult prediction scenarios.

Error Analysis and Resulting Improvements:

Error analysis was conducted by a team member outside of this notebook which motivated the feature engineering changes in Week 10. Based on that analysis, the model had more difficulty handling certain types of matchups where teams were closer in strength, which is expected.

Thus, we created new features to better capture these situations which include variables such as: absolute seed difference, efficiency margin differences, as well as indicator flags for toss-up and moderately competitive games. These additions provide the model with more structured information about matchup difficulty.

The identification of redundant variables and extreme values in the dataset led to the removal of highly correlated features and the implementation of quantile-based outlier clipping. These changes were aimed at improving overall data quality and model stability.

Model Refitting with Updated Data:

The model selected in Week 9 was an XGBoost classifier with tuned hyperparameters and a decision threshold of 0.45. The baseline model with Week 9 data achieved a validation accuracy of 0.7778, precision of 0.7600, and recall of 0.9500.

Metrics by variant:

	variant	dataset	accuracy	precision	recall
2	baseline	test	0.6719	0.7009	0.8824
0	baseline	train	0.9191	0.8966	0.9949
1	baseline	validation	0.7778	0.7600	0.9500
5	improvement1_prune	test	0.6953	0.7091	0.9176
3	improvement1_prune	train	0.9182	0.8946	0.9962
4	improvement1_prune	validation	0.7460	0.7400	0.9250
8	improvement2_clip	test	0.6953	0.7091	0.9176
6	improvement2_clip	train	0.9234	0.9008	0.9962
7	improvement2_clip	validation	0.7778	0.7600	0.9500
11	improvement3_flags	test	0.7109	0.7182	0.9294
9	improvement3_flags	train	0.9234	0.9008	0.9962
10	improvement3_flags	validation	0.7460	0.7400	0.9250

The first improvement (removal of near-duplicate features) resulted in a decrease in validation performance, with accuracy dropping to 0.7460 and recall to 0.9250. This suggests that some of the removed features, while highly correlated, still contained useful signals for the model.

The second improvement (quantile clipping) produced identical validation performance to the baseline model, with accuracy of 0.7778, precision of 0.7600, and recall of 0.9500. This indicates that clipping outliers helped stabilize the data without negatively affecting model performance, but did not provide any improvement.

The third improvement (engineered matchup features) also resulted in lower validation performance, with accuracy of 0.7460 and recall of 0.9250. While these features were intended to improve performance on closer matchups, they did not lead to better generalization.

Final Model Selection and Error Insights:

Based on validation performance, the baseline model from Week 9 is selected as the final model to be deployed. It achieved the highest validation accuracy and recall compared to all Week 10 variants.

Chosen variant: baseline

	variant	dataset	accuracy	precision	recall
0	baseline	train	0.9191	0.8966	0.9949
1	baseline	validation	0.7778	0.7600	0.9500
2	baseline	test	0.6719	0.7009	0.8824

There is a clear increase in error as we move from training to validation to test. The training dataset has the lowest error (0.0809), followed by the validation dataset (0.2222), and the test dataset has the highest error (0.3281) which is expected in machine learning models.

The increase in error from training to validation indicates that the model is overfitting as it performs significantly better on the training data than on unseen data. The test error also supports that claim as it struggles more with newer seasons. Even so, recall remains relatively high across all datasets which shows the model is consistently identifying wins. However, the drop in precision on the test set shows an increase in false positives.

